IP Addressing- I

Course Code: CSC 3116 Course Title: Computer Networks

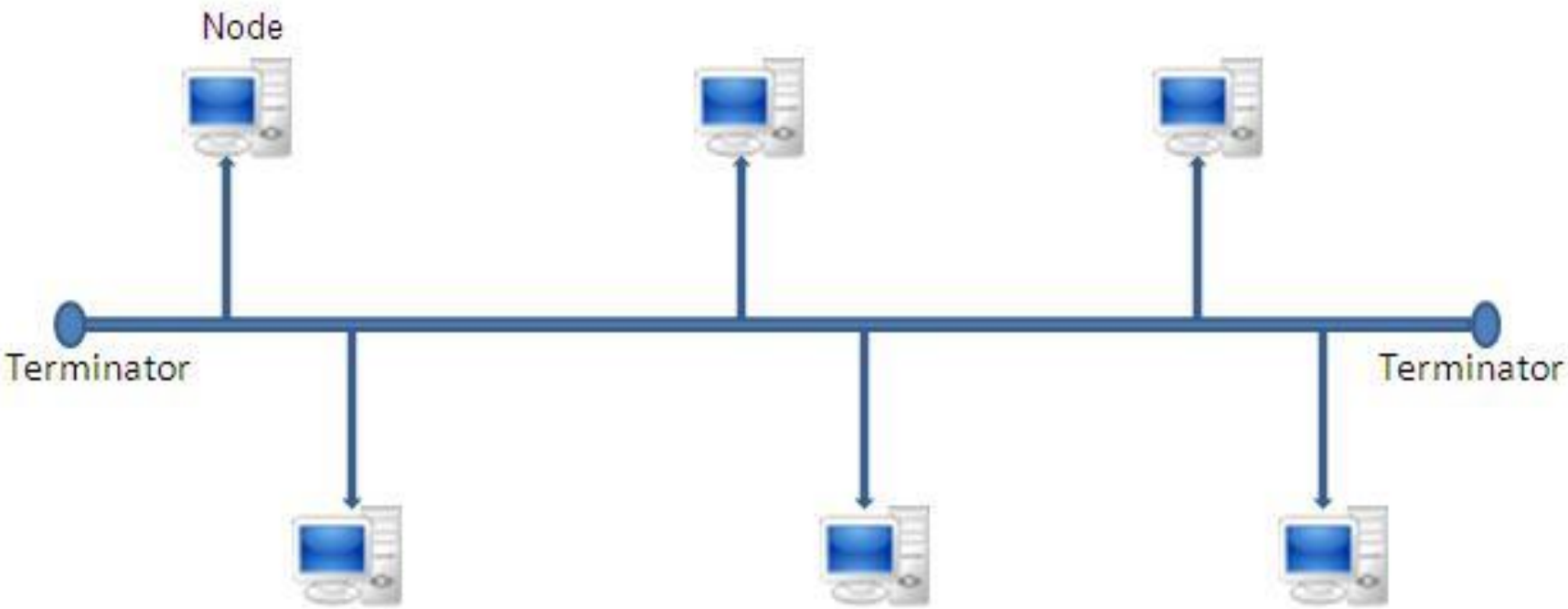
Dept. of Computer Science
Faculty of Science and Technology

Lecturer No:	Lab 1	**Week No:**	1	**Semester:**	Summer 2022-23
Lecturer:	*Sharifa Rania Mahmud; rania@aiub.edu*				

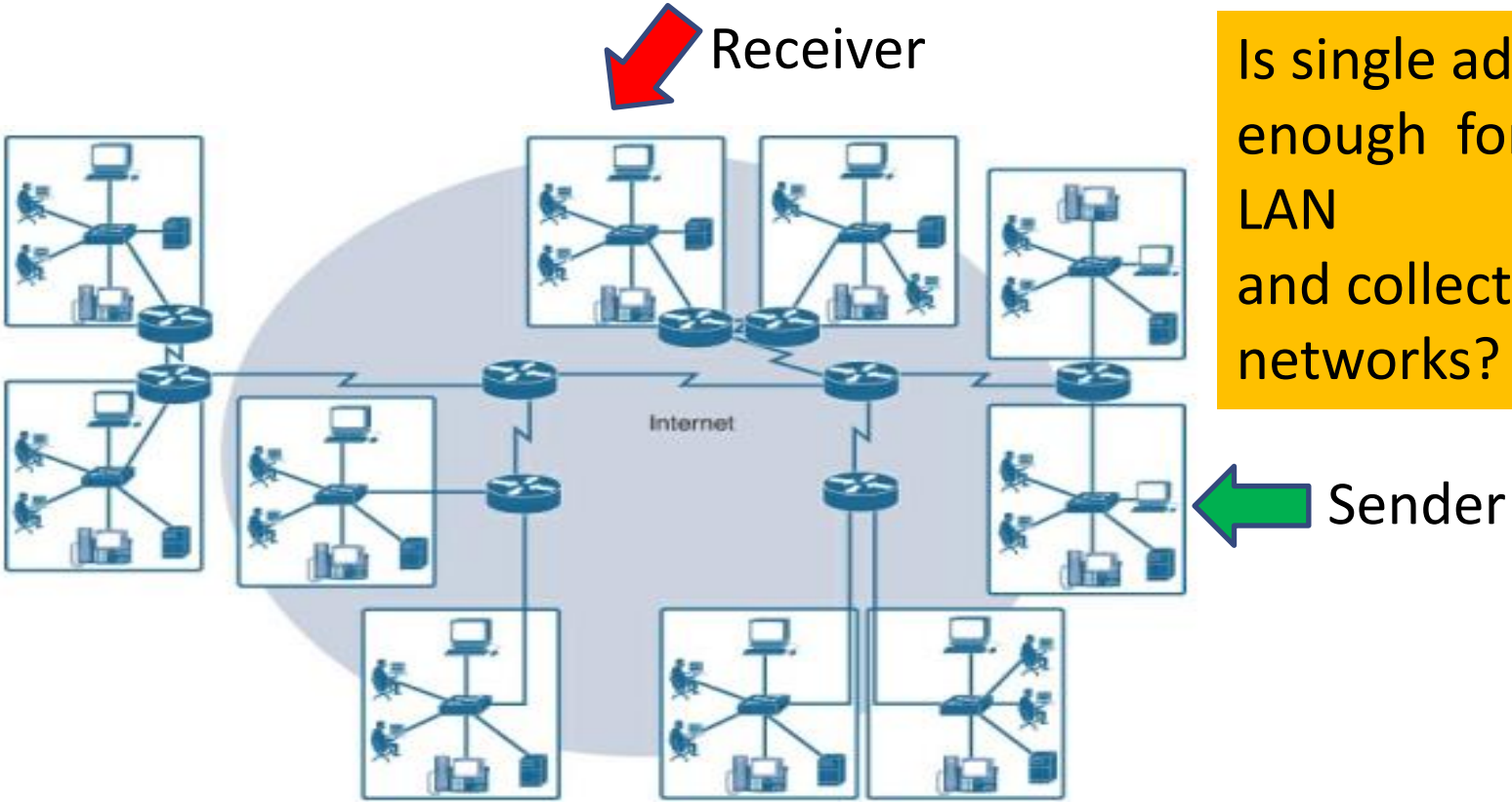
Lecture Outline

- Classification of Address
- Conversion from binary to decimal and vice-versa
- Range of Address

How to identify the recipient?



How to identify the recipient? (contd…)



Is single address is enough for both LAN
and collection of networks?

Classification of Address

- Address
  - Physical Address
  - Logical Address
  - Port Address

Physical Address

- the address of a node as defined by its LAN
- The lowest-level address
- The size and format of these addresses vary depending on the network
    - Ethernet uses 6 bytes address (imprinted on Network Interface Card (NIC))

Physical Address (contd…)

- 6 byte address is also called *Medium Access Control (MAC)* address
- No two NICs ever share the same MAC address
- Either imprinted on the surface or burnt into a ROM chip

d: Hexadecimal digit

$$d_1d_2 : d_3d_4 : d_5d_6 : d_7d_8 : d_9d_{10} : d_{11}d_{12}$$

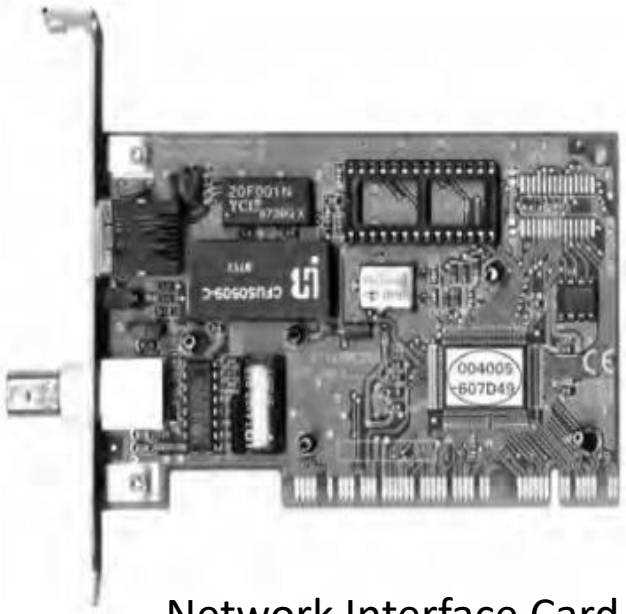
6 bytes = 12 hexadecimal digits = 48 bits

CA1885-0CB72C

CA-18-85-0C-B7-2C

Organizationally Unique Identifier

Device ID



Network Interface Card

Physical Address (contd…)

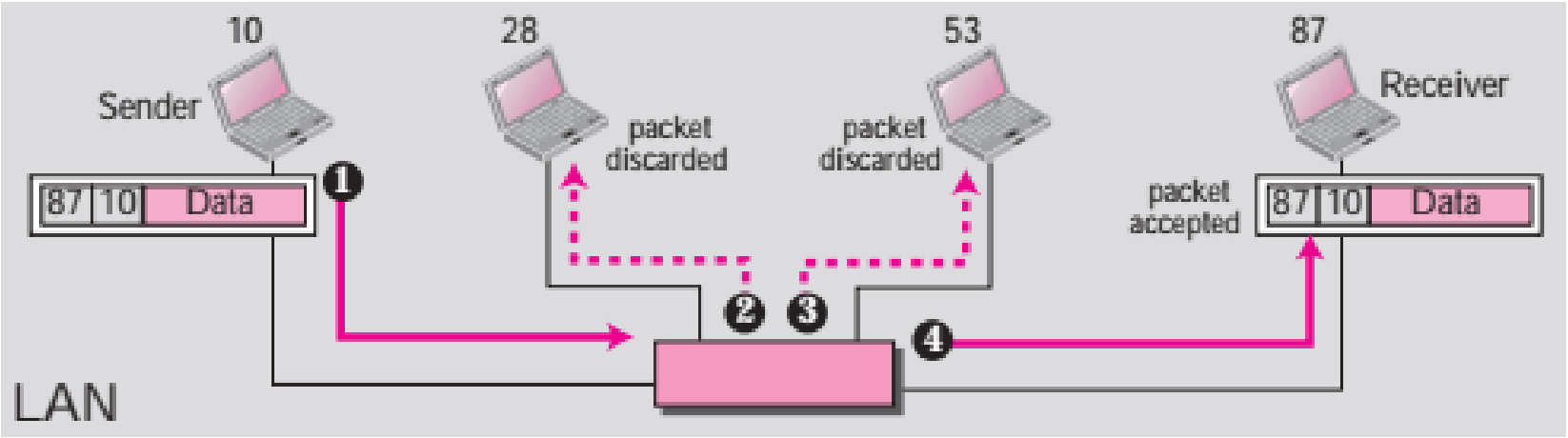
What's Your MAC Address?

You can readily determine your MAC address on a modern computer from the command line.

1. In Windows Vista/7, click Start, enter **`cmd`** in the Start Search text box, and press the ENTER key to get to a command prompt.
2. In Windows 8, simply type **`cmd`** at the Start screen and press ENTER when the Command Prompt option appears on the right.
3. At the command prompt, type the command **`ipconfig /all`** and press the ENTER key.

Physical Address (contd…)

- Use of Physical Address



Logical Address

- Physical address is not suitable for internetwork as different networks can have different address formats
- A universal addressing system is needed in which each host can be identified uniquely, regardless of the underlying physical network
- Can be changed depending on the network
- No two publicly addressed and visible hosts on the Internet can have the same logical address (widely known as Internet Protocol (IP) address.
- 32-bits length

Port Address

- Used to identify a process (Email, FTP)
- 16-bits length
- Internet Assigned Number Authority (IANA) assigns Port address

Port Address

Well-Known Port	Application	Protoc
20	File Transfer Protocol (FTP) Data	TCP
21	File Transfer Protocol (FTP) Control	TCP
23	Telnet	TCP
25	Simple Mail Transfer Protocol (SMTP)	TCP
69	Trivial File Transport Protocol (TFTP)	UDP
80	Hypertext Transfer Protocol (HTTP)	TCP
110	Post Office Protocol 3 (POP3)	TCP
194	Instant Relay Chat (IRC)	TCP
443	Secure HTTP (HTTPS)	TCP
520	Routing Information Protocol	UDP

Binary-to-Decimal Conversion

For addressing, we require
to convert a 8-bits binary to decimal
To convert a decimal number of up to 255 to binary number

Digit	x_8	x_7	x_6	x_5	x_4	x_3	x_2	x_1
Position of digit, *i*	8	7	6	5	4	3	2	1
Contribution of the digit, 2^{i-1}	2^{8-1}	2^{7-1}	2^{6-1}	2^{5-1}	2^{4-1}	2^{3-1}	2^{2-1}	2^{1-1}
Decimal contribution	128	64	32	16	8	4	2	1

Binary-to-Decimal Conversion

$$if \;\; x_i = 1, add\ 2^{i-1}\ to\ the\ sum \qquad for\ \forall i$$

Digits	**1**	**0**	**0**	**1**	**1**	**0**	**0**	**1**
Position	8	7	6	5	4	3	2	1
Contribution	128			16	8			1

128+16+8+1=153

$(10011001)_2=(153)_{10}$

Binary-to-Decimal Conversion

0	**1**	**0**	**0**	**1**	**1**	**0**	**1**

Digits	**0**	**1**	**0**	**0**	**1**	**1**	**0**	**1**
Position	8	7	6	5	4	3	2	1
Contribution		64			8	4		1

$$64+8+4+1=77$$

$$(01001101)_2=(77)_{10}$$

Decimal-to-Binary Conversion

❖$(172)_{10}$ =( )$_2$

$(172)_{10}$ =(10101100)$_2$

Step 1. Because 172 is NOT less than 128, place a 1 in the 128 position and subtract 128 (1 * 128).

Step 2. Because 44 is less than 64, place a 0 in the 64 position and subtract 0 (0 * 64).

Step 3. Because 44 is NOT less than 32, place a 1 in the 32 position and subtract 32 (1 * 32).

Step 4. Because 12 is less than 16, place a 0 in the 16 position and subtract 0 (0 * 16).

Step 5. Because 12 is NOT less than 8, place a 1 in the 8 position and subtract 8 (1 * 8).

Step 6. Because 4 is NOT less than 4, place a 1 in the 4 position and subtract 4 (1 * 4).

Step 7. Because 0 is less than 2, place a 0 in the 2 position and subtract 0 (0 * 2).

Step 8. Because 0 is less than 1, place a 0 in the 1 position and subtract 0 (0 * 1).

Decimal-to-Binary Conversion

❖$(20)_{10}$ =( )$_2$

$(20)_{10}$ =(00010100)$_2$

Step 1. Because 20 is less than 128, place a 0 in the 128 position and subtract 0 (0 * 128).

Step 2. Because 20 is less than 64, place a 0 in the 64 position and subtract 0 (0 * 64).

Step 3. Because 20 is less than 32, place a 0 in the 32 position and subtract 0 (0 * 32).

Step 4. Because 20 is NOT less than 16, place a 1 in the 16 position and subtract 16 (1 * 16).

Step 5. Because 4 is less than 8, place a 0 in the 8 position and subtract 0 (0 * 8).

Step 6. Because 4 is NOT less than 4, place a 1 in the 4 position and subtract 1 (1 * 4).

Step 7. Because 0 is less than 2, place a 0 in the 2 position and subtract 0 (0 * 2).

Step 8. Because 0 is less than 1, place a 0 in the 1 position and subtract 0 (0 * 1).

IP Address

- IP=Internet Protocol
- Unique and universal
- 32-bit binary address
- Divided into four octate
- Octets are separated by dot (.) sign

Address space
Total number of address used by a protocol
For b bits, the address space is 2^b

The address space of IPv4 is 2^{32} or 4,294,967,296.

10000000000010110000001100011111

10000000 00001011 00000011 00011111

Octet

IP Address 10000000.00001011.00000011.00011111

Doted Decimal of IP address 128.11.3.31

IP Address

Example

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Example

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

IP Address

Example

Find the error, if any, in the following IPv4 addresses:

a. 111.56.045.78

b. 221.34.7.8.20

Ranges of Address

Example

Find the number of addresses in a range if the first address is 146.102.29.0 and the last address is 146.102.32.255.

Solution

$$\textbf{Number of addresses} = (0 \times 256^3 + 0 \times 256^2 + 3 \times 256^1 + 255 \times 256^0) + 1 = 1024$$

Example

The first address in a range of addresses is 14.11.45.96. If the number of addresses in the range is 32, what is the last address?

Solution

$$\textbf{Last address} = (14.11.45.96 + 0.0.0.31)_{256} = 14.11.45.127$$

References

1. Official Cert Guide CCNA 200-301 , vol. 1, *W. Odom*, Cisco Press, First Edition, 2019, USA.

2. CCNA Routing and Switching, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.

3. Cisco IOS Configuration Fundamentals Command Reference. http://www.cisco.com

Books

1. Official Cert Guide CCNA 200-301 , vol. 1, *W. Odom*, Cisco Press, First Edition, 2019, USA.

2. CCNA Routing and Switching, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.